Supplementary Material

Search Terms

MEDLINE

((((((((((((("Physical Fitness"[Mesh]) OR "Physical Functional Performance"[Mesh]) OR ("Range of Motion, Articular"[Mesh] OR "Mobility Limitation"[Mesh])) OR ("Muscle Strength"[Mesh] OR "Muscle Strength Dynamometer"[Mesh] OR "Hand Strength"[Mesh] OR "Resistance Training"[Mesh] OR "Circuit-Based Exercise"[Mesh])) OR "Activities of Daily Living"[Mesh]) OR ("Gait"[Mesh] OR "Walking Speed"[Mesh]))) OR ("Physical function"[TW] OR "physical performance"[TW] OR "physical fitness"[TW] OR endurance[TW] OR “oxygen uptake”[TW] OR “maximal oxygen uptake”[TW] OR “peak oxygen uptake”[TW] OR “VO2 max”[TW] OR “VO2 peak”[TW] OR “submaximal exercise test”[TW] OR “functional performance”[TW] OR fitbit[TW] OR walking[TW] OR step[TW] OR mobility[TW] OR strength[TW] OR "6 minute walk test"[TW] OR "6 MWT"[TW] OR 6MWT[TW] OR "walking test"[TW] OR "Intermittent shuttle walk test"[TW] OR ISWT[TW] OR "Gait speed"[TW] OR "Stair climb power test"[TW] OR "Repeated chair stands"[TW] OR "chair stands"[TW] OR “sit to stand”[TW] OR “sit-to-stand”[TW] OR "Timed up and go"[TW] OR "TUG test"[TW] OR TUG[TW] OR "Walk-stair climb test"[TW] OR “walk stair climb test”[TW] OR “activity questionnaire”[TW] OR "Katz activities of daily living"[TW] OR "Katz ADL"[TW] OR "activities of daily living"[TW] OR "Lawton activities of daily living"[TW] OR "functional performance"[TW] OR "walking"[TW] OR "Lawton ADL"[TW]))) AND (((((((("Kidney Diseases"[Mesh]) OR "Renal Insufficiency, Chronic"[Mesh]) OR "Kidney Transplantation"[Mesh]) OR ("Dialysis"[Mesh] OR "Renal Dialysis"[Mesh] OR "Peritoneal Dialysis"[Mesh] OR "Hemodialysis, Home"[Mesh])) OR ("Kidney Failure, Chronic"[Mesh] OR "Renal Failure, Progressive, with Hypertension" [Supplementary Concept]))) OR ("Kidney disease"[TW] OR "chronic kidney disease"[TW] OR CKD[TW] OR "end stage kidney disease"[TW] OR ESKD[TW] OR "renal disease"[TW] OR "chronic renal disease"[TW] OR "chronic renal failure"[TW] OR "renal failure"[TW] OR "end stage renal failure"[TW] OR ESRF[TW] OR Dialysis[TW] OR haemodialysis[TW] OR hemodialysis[TW] OR "peritoneal dialysis"[TW] OR "Kidney transplant"[TW] OR "kidney transplantation"[TW] OR "renal transplant"[TW] OR "renal transplantation"[TW])))))) NOT (("Animals"[Mesh]) NOT "Humans"[Mesh])))

Embase

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MWT":TI,AB OR 6MWT:TI,AB OR "walking test":TI,AB OR "Intermittent shuttle walk test":TI,AB OR ISWT:TI,AB OR "Gait speed":TI,AB OR "Stair climb power test":TI,AB OR "Repeated chair stands":TI,AB OR "chair stands":TI,AB OR “sit to stand”:TI,AB OR “sit-to-stand”:TI,AB OR "Timed up and go":TI,AB OR "TUG test":TI,AB OR TUG:TI,AB OR "Walk-stair climb test":TI,AB OR “walk stair climb test”:TI,AB OR “activity questionnaire”:TI,AB OR "Katz activities of daily living":TI,AB OR "Katz ADL":TI,AB OR "activities of daily living":TI,AB OR "Lawton activities of daily living":TI,AB OR "functional performance":TI,AB OR "walking":TI,AB OR "Lawton ADL":TI,AB))) AND (((((("Kidney Diseases"[Mesh]) OR "Renal Insufficiency, Chronic"[Mesh]) OR "Kidney Transplantation"[Mesh]) OR ("Dialysis"[Mesh] OR "Renal Dialysis"[Mesh] OR "Peritoneal Dialysis"[Mesh] OR "Hemodialysis, Home"[Mesh])) OR ("Kidney Failure, Chronic"[Mesh] OR "Renal Failure, Progressive, with Hypertension" [Supplementary Concept]))) OR ("Kidney disease"[TIAB] OR "chronic kidney disease":TI,AB OR “renal insufficiency” :TI,AB OR CKD:TI,AB OR "end stage kidney disease":TI,AB OR ESKD:TI,AB OR "renal disease":TI,AB OR "chronic renal disease":TI,AB OR "chronic renal failure":TI,AB OR "renal failure":TI,AB OR "end stage renal failure":TI,AB OR ESRF:TI,AB OR Dialysis:TI,AB OR haemodialysis:TI,AB OR hemodialysis:TI,AB OR "peritoneal dialysis":TI,AB OR "Kidney transplant":TI,AB OR "kidney transplantation":TI,AB OR "renal transplant":TI,AB OR "renal transplantation":TI,AB))))) NOT (("Animals"[Mesh]) NOT "Humans"[Mesh])))

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1. Chronic kidney disease*
2. Hemodialysis*
3. Peritoneal dialysis
4. Kidney transplant*
5. Physical fitness*

Table S1 – Summary of intervention type, fitness outcome and assessment method across the studies.

Author	Year	Country	Journal	Cohort	Intervention Type	Fitness Outcome	Assessment Method
Tawney et al. [1]	2000	USA	American Journal of Kidney Diseases	HD	Lifestyle counselling	Physical function	KDQOL-SF SF-36
Castaneda et al. [2]	2001	USA	Annals of Internal Medicine	CKD	Resistance Low protein diet	Muscle CSA Strength BMI Peak oxygen consumption	CT, Histology 1-RM BMI VO_2 peak
Brass et al. [3]	2001	USA	American Journal of Kidney Diseases	HD	L-carnitine	Exercise capacity	VO_2 max
Konstantinidou et al. [4]	2002	Greece	Journal of Rehabilitation Medicine	HD	Aerobic	Aerobic capacity	VO_2 peak HR max Anaerobic threshold Lactate threshold
Painter et al. [5]	2002	USA	Transplantation	KTx	Aerobic	Cardiorespiratory fitness Strength Muscle mass HRQOL	VO_2 peak Quadriceps strength DEXA SF-36
Koufaki et al. [6]	2002	UK	Clinical Physiology & Functional Imaging	HD, PD	Aerobic	Exercise capacity Functional capacity	VO_2 peak 5 STS 60s STS NSRI
De Paul et al. [7]	2002	Canada	American Journal of Kidney Diseases	HD	Aerobic Resistance	Exercise capacity Strength	Graded ergometer (time on test) 5-RM

						HRQOL Walking distance	SF-36 6MWT
Painter et al. [8]	2002	USA	American Journal of Kidney Diseases	HD	Aerobic ESA	Exercise capacity HRQOL	VO_2 peak Anaerobic threshold SF-36
Painter et al. [9]	2003	USA	Kidney International	KTx	Steroid withdrawal	Cardiorespiratory fitness Strength Muscle mass Fat mass	VO_2 peak HR max Ventilatory threshold Quadriceps strength DEXA DEXA
Painter et al. [10]	2003	USA	American Journal of Kidney Diseases	KTx	Aerobic	Exercise capacity BMI Lipids	Peak METS BMI Lipids
Castaneda et al. [11]	2004	USA	American Journal of Kidney Diseases	CKD	Resistance Low protein diet	Body composition Strength	Muscle CSA BMI 1-RM
Kouidi et al. [12]	2004	Greece	Clinical Nephrology	HD	Aerobic	Physical fitness	VO_2 peak HR max Anaerobic threshold Lactate threshold
Molsted et al. [13]	2004	Denmark	Nephron Clinical Practice	HD	Aerobic Resistance	Aerobic capacity Physical function	VO_2 peak 2-min stair climb Squat test SF-36
Parsons et al. [14]	2004	Canada	Clinical Nephrology	HD	Aerobic	Work performance QOL	Graded ergometer (peak power) SF-36

Van Vilsteren et al. [15]	2005	Netherlands	Nephrology Dialysis Transplantation	HD	Aerobic Resistance	Exercise capacity HRQOL Strength	VO_2 peak SF-36 10 STS
Johansen et al. [16]	2006	USA	Journal of the American Society of Nephrology	HD	Resistance Nandrolone	Lean muscle Fat mass Muscle CSA Strength Physical performance Functioning	DEXA DEXA MRI Quadriceps strength 20ft walk Stair climbing 5 STS SF-36
Cheema et al. [17]	2007	Australia	American Journal of Kidney Diseases	HD	Resistance	Muscle CSA Strength Exercise capacity Body composition	CT 1-RM 6MWT Muscle circumference
Cheema et al. [18]	2007	Australia	Journal of the American Society of Nephrology	HD	Resistance	Muscle CSA Strength Function Vitality Body composition Fat	CT 1-RM 6MWT SF-36 Muscle circumference Skinfold thickness
Petraki et al. [19]	2008	Greece	Clinical Nephrology	HD	Resistance Flexibility	Baroreflex sensitivity Cardiorespiratory fitness	Arterial baroreflex VO_2 peak Graded treadmill (time on test)
Kouidi et al. [20]	2009	Greece	American Journal of Kidney Diseases	HD	Aerobic Resistance	Aerobic capacity HR variability	VO_2 peak Holter
Leehey et al. [21]	2009	USA	Cardiovascular Diabetology	CKD	Aerobic	Exercise Fat mass	VO_2 peak Plethysmography

						Lean muscle	Plethysmography
Segura-orti et al. [22]	2009	Spain	Clinical Nephrology	HD	Aerobic Resistance	Physical performance HRQOL Cardiorespiratory fitness Strength	10 STS 60s STS 6MWT SF-36 Graded treadmill (time on test, METS) Quadriceps strength
Ouzouni et al. [23]	2009	Greece	Clinical Rehabilitation	HD	Aerobic Resistance Flexibility	Cardiorespiratory fitness QOL	VO_2 peak SF-36 Spitzer index
Wilund et al. [24]	2010	USA	Nephrology Dialysis Transplantation	HD	Aerobic	Physical performance Epicardial fat	ISWT TTE
Kouidi et al. [25]	2010	Greece	European Journal of Cardiovascular Prevention and Rehabilitation	HD	Aerobic Resistance	Physical capacity Cardiac	VO_2 peak HR variability
Chen et al. [26]	2010	USA	Nephrology Dialysis Transplantation	HD	Resistance	Physical performance Strength QOL Muscle mass Fat mass	*SPPB Quadriceps strength SF-36 DEXA DEXA
Koh et al. [27]	2010	Australia	American Journal of Kidney Diseases	HD	Aerobic	Physical function Arterial stiffness	6MWT TUG SF-36 HGS Pulse wave velocity

Dobsak et al. [28]	2011	Czech Republic	Artificial Organs	HD	Aerobic Electromyostimulation	Walk distance Power QOL	6MWT Quadriceps strength SF-36
Bullani et al. [29]	2011	Switzerland	Journal of Renal Nutrition	HD	Resistance	Physical function	Tinetti instrument TUG 1 leg balance test 6MWT
Dong et al. [30]	2011	USA	Journal of Renal Nutrition	HD	Resistance Diet	Muscle mass Fat mass Strength Body composition	DEXA DEXA 1-RM BIA
Muirhead et al. [31]	2011	Canada	Hemodialysis International	HD	ESA	Exercise tolerance Physical function	Graded treadmill (time on test) 6MWT KDQOL-SF SIP
Mustata et al. [32]	2011	Canada	International Urology and Nephrology	CKD	Aerobic	Exercise capacity HRQOL Arterial stiffness	VO_2 peak Graded ergometer (time on test) SF-36 Pulse wave velocity
Reboredo et al. [33]	2011	Brazil	Archives of Physical Medicine and Rehabilitation	HD	Aerobic	Exercise tolerance Endurance	VO_2 peak Anaerobic threshold Graded ergometer (time on test)
Gregory et al. [34]	2011	USA	Growth Hormone and IGF Research	CKD	Aerobic Resistance Flexibility	Physical performance Nutrition	VO_2 peak Graded treadmill BMI
Hall et al. [35]	2012	USA	Clinical Journal of the American	HD	Dialysis frequency	Physical performance Physical health	*SPPB SF-36

			Society of Nephrology			Physical function	SF-36
Orcy et al. [36]	2012	Brazil	Physiotherapy Research International	HD	Aerobic Resistance	Walking capacity	6MWT
Song et al. [37]	2012	Korea	Journal of Korean Academy of Nursing	HD	Resistance	Muscle mass Physical fitness QOL Lipids	BIA HGS Quad strength SF-36 Lipids
Giannaki et al. [38]	2013	Greece	Nephrology Dialysis Transplantation	HD	Aerobic Resistance	Functional capacity	NSRI 5 STS 60s STS
Kouidi et al. [39]	2013	Greece	Nephrology Dialysis Transplantation	KTx	Aerobic Resistance	CV function	VO_2 peak HR variability
De Lima et al. [40]	2013	Brazil	Renal Failure	HD	Aerobic Resistance	Respiratory strength Respiratory function QOL Functional capacity	MIP/MEP Spirometry KDQOL-SF 4 min step test
Giannaki et al. [41]	2013	Greece	BMC Nephrology	HD	Aerobic Ropinorole	Physical performance QOL Muscle mass Fat mass Muscle size	NSRI 5 STS 30s STS 60s STS Normal walk test Fast walk test SF-36 DEXA DEXA CT

Howden et al. [42]	2013	Australia	Clinical Journal of the American Society of Nephrology	CKD	Aerobic Resistance	Cardiorespiratory fitness Body composition Arterial stiffness	VO_2 peak BMI Pulse wave velocity
Molsted et al. [43]	2013	Denmark	Journal of Renal Nutrition	HD, PD	Resistance	Physical performance Power Strength HRQOL Muscle morphology	30s STS Leg extensor rig Quadriceps strength SF-36 Histology
Pellizzaro et al. [44]	2013	Brazil	Renal Failure	HD	Resistance Respiratory training	Respiratory strength Respiratory function QOL Functional capacity	MIP/MEP Spirometry KDQOL-SF 6MWT
Supasyndh et al. [45]	2013	Thailand	Clinical Journal of the American Society of Nephrology	HD	Anabolic steroid	Muscle mass Fat mass Strength QOL Muscle morphology	DEXA DEXA HGS SF-36 Histology
Teng et al. [46]	2013	Taiwan	Western Journal of Nursing Research	CKD	Lifestyle modification	Body composition Exercise capacity	BMI Waist to hip ratio 6MWT
Headley et al. [47]	2014	USA	American Journal of Kidney Diseases	CKD	Aerobic	Arterial stiffness Aerobic capacity HRQOL	Pulse wave velocity VO_2 peak SF-36
Tzvetanov et al. [48]	2014	USA	Transplant Proceedings	KTx	Resistance	BMI QOL Arterial stiffness Strength	BMI SF-36 Pulse wave velocity 1-RM

Baria et al. [49]	2014	Brazil	Nephrology Dialysis Transplantation	CKD	Aerobic	Visceral fat Fat mass Muscle mass BMI Physical capacity Functional capacity	CT DEXA Waist to hip ratio DEXA BMI VO_2 peak 6MWT 30s STS
Bohm et al. [50]	2014	Canada	Nephrology Dialysis Transplantation	HD	Aerobic	Aerobic capacity Strength QOL Flexibility	VO_2 peak 6MWT 30s STS SF-36 Sit and reach test
Kirkman et al. [51]	2014	UK	Journal of Cachexia, Sarcopenia and Muscle	HD	Resistance	Muscle CSA Strength Balance QOL Aerobic capacity	MRI Quadriceps strength 30s STS TUG SF-36 6MWT
Rossi et al. [52]	2014	USA	Clinical Journal of the American Society of Nephrology	CKD	Aerobic Resistance	Physical function HRQOL	6MWT 10 STS Gait speed SF-36
Wu et al. [53]	2014	China	Journal of International Medical Research	HD	Aerobic	Exercise capacity HRQOL	6MWT Stair climbing 60s STS HGS 10 STS KDQOL-SF

Aoike et al. [54]	2015	Brazil	International Urology and Nephrology	CKD	Aerobic	Cardiorespiratory fitness Functional capacity	VO_2 peak 6MWT 2-min step test 30s STS Arm curl test Sit and reach test TUG
Howden et al. [55]	2015	Australia	American Journal of Kidney Diseases	CKD	Aerobic Resistance	Exercise capacity Power Strength Body composition	Peak METS 6MWT TUG HGS BMI
Manfredini et al. [56]	2015	Italy	American Journal of Nephrology	HD	Aerobic	Muscle characteristics Vascular function Functional capacity	Resting muscle VO_2 Toe-brachial index Arterial stiffness index 6MWT
Reboredo et al. [57]	2015	Brazil	European Journal of Preventive Cardiology	HD	Aerobic	Exercise tolerance	VO_2 peak
Tao et al. [58]	2015	Hong Kong	International Journal of Nursing Studies	HD	Resistance Flexibility	Physical function QOL ADLs	Gait speed SF-36 10 STS
Watson et al. [59]	2015	UK	American Journal of Kidney Diseases	CKD	Aerobic Resistance	Muscle CSA Strength Exercise capacity	US Quadriceps strength ISWT
Wu et al. [60]	2015	USA	Journal of Medicinal Food	HD	Pomegranate	Body composition Arterial structure	BMI US Pulse wave velocity CIMT

						Cardiac Physical performance Strength	TTE Gait speed ISWT Chair stands Arm curl test Hamstring flexibility Shoulder flexibility TUG Quadriceps strength
Greenwood et al. [61]	2015	UK	American Journal of Kidney Diseases	CKD	Aerobic Resistance	Cardiorespiratory fitness Vascular function Body composition	VO_2 peak Pulse wave velocity BMI
Greenwood et al. [62]	2015	UK	American Journal of Kidney Diseases	KTx	Aerobic Resistance	Cardiorespiratory fitness Vascular function Strength Functional capacity QOL Body composition	VO_2 peak HR max Ventilatory threshold Pulse wave velocity Quadriceps strength 60s STS SF-36 BMI
Groussard et al. [63]	2015	France	Applied Physiology, Nutrition and Metabolism	HD	Aerobic	Physical fitness Aerobic capacity Lipids Body composition Fat mass Muscle mass	VO_2 peak 6MWT Lipids BMI DEXA DEXA
Matsufuji et al. [64]	2015	Japan	Journal of Renal Nutrition	HD	Resistance	ADLs QOL	FIM SF-36

						Muscle size Strength Walking distance Chair stands	Muscle circumference Quadriceps strength 6MWT Chair stands
Van Craenenbroeck et al. [65]	2015	Belgium	American Journal of Kidney Diseases	CKD	Aerobic	Endothelial function Aerobic capacity HRQOL Arterial stiffness	Arterial FMD VO_2 peak KDQOL-SF Pulse wave velocity
Karelis et al. [66]	2016	Canada	Canadian Journal of Diabetes	KTx	Resistance	Glycemic Cardiorespiratory fitness Fat mass Strength Lipids	OGTT VO_2 peak DEXA BMI Waist to hip ratio 1-RM Lipids
Leehey et al. [67]	2016	USA	American Journal of Nephrology	CKD	Aerobic Resistance Diet	Physical function HRQOL Fat mass	Graded treadmill VO_2 peak 6MWT TUG Quadriceps strength SF-36 Plethysmography
Liao et al. [68]	2016	Taiwan	Medicine	HD	Aerobic	BMD ADLs Body composition	DEXA 6MWT BMI
Thompson et al. [69]	2016	Canada	British Medical Journal Open	HD	Aerobic Resistance Stretching	QOL Physical performance Strength	SF-36 *SPPB 6MWT 30s STS

							1-RM
Pomidori et al. [70]	2016	Italy	Journal of Nephrology	HD	Aerobic	Exercise capacity Respiratory function	6MWT Spirometry
Bennett et al. [71]	2016	Australia	Nephrology Dialysis Transplantation	HD	Resistance	Physical function QOL	30s STS TUG FSST KDQOL-SF
Gokbel et al. [72]	2016	Turkey	American Journal of Therapeutics	HD	Coenzyme Q10	Exercise performance	6MWT VO_2 max
Hristea et al. [73]	2016	France	Nephrology	HD	Aerobic Resistance	Muscle mass Physical capacity QOL Strength	BIA 6MWT SF-36 Quadriceps strength
Martin-Alemany et al. [74]	2016	Mexico	Nephrology Dialysis Transplantation	HD	Resistance Diet	Body composition Fat mass Strength QOL	BIA BMI Skinfold thickness HGS KDQOL-SF
Neto et al. [75]	2016	Brazil	The Journal of Physical Therapy Science	HD	Resistance Respiratory training	Respiratory strength Respiratory function Strength	MIP/MEP Spirometry HGS
Olvera-Soto et al. [76]	2016	Mexico	Journal of Renal Nutrition	HD	Resistance	Body composition Fat mass Strength	BMI Skinfold thickness HGS
Roxo et al. [77]	2016	Brazil	Brazilian Journal of Nephrology	HD	Electromyostimulation	Respiratory function Respiratory strength Functional capacity	Spirometry MIP/MEP 1-RM 6MWT

Anastasia et al. [78]	2016	Greece	Journal of Clinical and Experimental Nephrology	HD	Aerobic	Physical capacity HRQOL	6MWT 10 STS Sit and reach test HGS TUG SF-36
Manfredini et al. [79]	2017	Italy	Journal of the American Society of Nephrology	HD, PD	Aerobic	Physical performance QOL	6MWT 5 STS KDQOL-SF
O'Connor et al. [80]	2017	UK	PLOS One	KTx	Aerobic Resistance	Arterial stiffness Cardiorespiratory fitness	Pulse wave velocity VO_2 peak
Onofre et al. [81]	2017	Brazil	Brazilian Journal of Nephrology	KTx	Aerobic Resistance	Walking capacity Respiratory strength Respiratory function Strength	6MWT MIP/MEP Spirometry HGS Quadriceps strength
Bruggemann et al. [82]	2017	Brazil	Archives of Physical Medicine and Rehabilitation	HD	Electromyostimulation	Strength Exercise capacity	Quadriceps strength 6MWT
Frih et al. [83]	2017	Tunisia	Frontiers in Physiology	HD	Aerobic Resistance	Physical capacity HRQOL	10 STS 60s STS HGS TUG 6MWT SF-36
Frih et al. [84]	2017	Tunisia	Libyan Journal of Medicine	HD	Aerobic Resistance	Functional capacity	TUG 6MWT

					Q'uran citation	HRQOL	SF-36
Gomes et al. [85]	2017	Brazil	Journal of Renal Nutrition	CKD	Aerobic	Physical capacity Functional capacity Body composition Fat mass Muscle mass Strength	VO_2 peak 6MWT TUG 30s STS BMI DEXA DEXA HGS
Headley et al. [86]	2017	USA	Nephrology	CKD	Aerobic	Aerobic capacity	VO_2 peak Anaerobic threshold
Hiraki et al. [87]	2017	Japan	BMC Nephrology	CKD	Aerobic Resistance	Strength	HGS Quadriceps strength
Sarmento et al. [88]	2017	Brazil	Clinical Rehabilitation	KTx, CKD	Resistance Balance Stretching	Function Exercise capacity Respiratory strength	Barthel index Incremental step test MIP/MEP
Schardong et al. [89]	2017	Brazil	Artificial Organs	HD	Electromyostimulation	Strength Muscle CSA Functional capacity Endothelial function	Quadriceps strength 30s STS US 6MWT Arterial FMD
Tang et al. [90]	2017	China	International Journal of Nursing Practice	CKD	Aerobic	Physical function HRQOL	6MWT 10 STS KDQOL-SF
Barcellos et al. [91]	2018	Brazil	Journal of Human Hypertension	CKD	Aerobic Resistance	Functional capacity Lipids HRQOL	30s STS 2-min step test TUG Lipids SF-36

Alp Ikizler et al. [92]	2018	USA	Journal of the American Society of Nephrology	CKD	Aerobic Diet	Fat mass Muscle mass Cardiorespiratory fitness	DEXA BIA DEXA BIA VO_2 peak
Aoike et al. [93]	2018	Brazil	Clinical and Experimental Nephrology	CKD	Aerobic	Cardiorespiratpry fitness Functional capacity QOL	VO_2 peak 6MWT 2-min step test 30s STS Arm curl test Sit and reach test TUG SF-36
Cooke et al. [94]	2018	Canada	American Journal of Hypertension	HD	Aerobic	Arterial stiffness Body composition Function	Augmentation index Pulse wave velocity BMI Waist to hip ratio Gait speed HGS
Frih et al. [95]	2018	Tunisia	Disability and Rehabilitation	HD	Resistance Balance training	Balance Functional capacity	TUG 1 leg balance test Activities-specific balance test Tinetti instrument Berg balance scale Mini-balance evaluation 6MWT

McGregor et al. [96]	2018	UK	PLOS One	HD	Aerobic Electromyostimulation	Cardiorespiratory fitness Strength Cardiac Arterial stiffness	VO_2 peak Quadriceps strength TTE Pulse wave velocity
Segura-orti et al. [97]	2018	Spain	Scandinavian Journal of Medicine & Science in Sports	HD	Aerobic Resistance Virtual reality	Physical function	60s STS 10 STS 5 STS 6MWT Heel rises Gait speed
Stringuetta Belik et al. [98]	2018	Brazil	Nephron Clinical Practice	HD	Aerobic Stretching	Physical function	VO_2 peak
Watson et al. [99]	2018	UK	American Journal of Physiology - Renal Physiology	CKD	Aerobic Resistance	Strength Muscle CSA Exercise capacity Walking capacity	5-RM MRI US VO_2 peak ISWT
Barbosa et al. [100]	2018	Brazil	The Journal of Vascular Access	CKD	Resistance Blood flow restriction	Vascular measures Muscle size Strength	US Muscle circumference HGS
Medeiros et al. [101]	2018	Brazil	Disability and Rehabilitation	HD	Respiratory training	Respiratory strength Respiratory function Functional capacity QOL	MIP/MEP Spirometry 6MWT KDQOL-SF
Suzuki et al. [102]	2018	Japan	Artificial Organs	HD	Electromyostimulation	Strength Muscle CSA Physical function QOL	Quadriceps strength MRI TUG SF-8

Baggetta et al. [103]	2018	Italy	BMC Geriatrics	HD	Aerobic	Physical performance QOL	6MWT 5 STS KDQOL-SF
Beetham et al. [104]	2018	Australia	BMC Nephrology	CKD	Aerobic Resistance	Strength Cardiorespiratory fitness Exercise capacity Muscle mass	HGS VO_2 peak 6MWT Peak METS DEXA
Figueiredo et al. [105]	2018	Brazil	PLOS One	HD	Aerobic Respiratory training	Respiratory strength Functional capacity Strength HRQOL	MIP/MEP ISWT 30s STS KDQOL-SF
Hellberg et al. [106]	2018	Sweden	PLOS One	CKD	Aerobic Resistance Balance	Endurance Strength endurance Strength Balance	6MWT Stair climbing 30s STS Heel rises Toe lifts Quadriceps strength HGS Functional reach Berg balance scale
Macagnan et al. [107]	2018	Brazil	Lasers in Medical Science	HD	Photobiomodulation therapy	Strength	HGS
Rosa et al. [108]	2018	Brazil	Clinical Rehabilitation	HD	Resistance	Body composition Fat mass Muscle mass Strength Physical capacity	BMI DEXA DEXA HGS 30s STS

						Flexibility QOL	6MWT Sit and reach test SF-36
Tayebi et al. [109]	2018	Iran	Nephro-Urology Monthly	HD	Resistance	Endurance Strength	Leg extensor endurance Forearm endurance HGS
Uchiyama et al. [110]	2019	Japan	Scientific Reports	PD	Aerobic Resistance	Physical function Body composition HRQOL	ISWT HGS 1-RM BMI KDQOL-SF
Fuzari et al. [111]	2019	Brazil	Physiotherapy Theory and Practice	CKD, HD	Whole body vibration	Strength Muscle CSA Walking distance Balance QOL	Quadriceps strength US 6MWT Static GSI Dynamic GSI SF-36

Abbreviations: HD, hemodialysis; PD, peritoneal dialysis; KTx, kidney transplant; CKD, chronic kidney disease; Kidney disease quality of life (KDQOL), Short form 36 (SF-36), Short-form 8 (SF-8), Spitzer index, Sickness impact profile (SIP); HGS, hand grip strength; RM, repetition maximum; ADL, activities of daily living; FIM, functional independence measure; TUG, timed up and go; Mini-BEST, mini balance evaluation systems test; GSI, global stability index; 30s STS, 30 second sit to stand; 60s STS, 60 second sit to stand; 10 STS, 10 times sit to stand; 5 STS, 5 times sit to stand; FMD, flow-mediated dilation; CIMT, carotid intima-media thickness; HbA1c, glycosylated haemoglobin; OGTT, oral glucose tolerance test; VO_2, peak oxygen uptake; TTE, transthoracic echocardiography; HR, heart rate; 6MWT, 6-minute walk test; NSRI, North Staffordshire Royal Infirmary; ISWT, incremental shuttle walking test; METS, metabolic equivalents; BMI, body mass index; DEXA, dual x-ray absorptiometry; BIA, bioimpedance analysis; CT, computed tomography; CSA, cross-sectional area; US, ultrasound; MRI, magnetic resonance imaging; BMD, bone mineral density. * SPPB, short physical performance battery (not included in domain-specific analyses)

Table S2 – Distribution of tests/measurements across the stages of kidney disease.

HD	%	CKD	%	KTx	%	PD	%
6MWT	11.26	VO2 peak	12.98	VO2 peak	13.95	KDQOL-SF	11.76
SF-36	9.93	6MWT	9.16	BMI	9.30	5 STS	11.76
VO2 peak/max	5.96	BMI	6.87	DEXA	9.30	VO2 peak	5.88
Quad strength	4.97	SF-36	5.34	SF-36	6.98	6MWT	5.88
DEXA	4.97	DEXA	5.34	Pulse wave velocity	6.98	BMI	5.88
HGS	4.64	30s STS	4.58	Quad strength	6.98	SF-36	5.88
KDQOL-SF	3.97	HGS	4.58	1-RM	4.65	30s STS	5.88
30s STS	3.64	TUG	4.58	Lipids	4.65	HGS	5.88
TUG	3.31	Quad strength	3.82	MIP/MEP	4.65	Quad strength	5.88
5 STS	2.32	Pulse wave velocity	3.82	HR Max	4.65	1-RM	5.88
BMI	2.32	US	3.05	Ventilatory threshold	4.65	ISWT	5.88
10 STS	2.32	2-min step test	2.29	6MWT	2.33	Histology	5.88
MIP/MEP	1.99	10 STS	1.53	HGS	2.33	60s STS	5.88
Spirometry	1.99	1-RM	1.53	Peak METS	2.33	NSRI	5.88
60s STS	1.99	Arm curl test	1.53	Waist to hip ratio	2.33	Leg extensor power rig	5.88
1-RM	1.66	BIA	1.53	Barthel Index	2.33		
Graded exercise treadmill (time on test)	1.66	CT	1.53	Incremental step test	2.33		
Pulse wave velocity	1.32	Graded exercise treadmill (time on test)	1.53	HR variability	2.33		
Gait speed	1.32	ISWT	1.53	Spirometry	2.33		
BIA	1.32	KDQOL-SF	1.53	60s STS	2.33		
Anaerobic threshold	1.32	Peak METS	1.53	OGTT	2.33		
CT	0.99	Plethysmography	1.53				
ISWT	0.99	Sit and reach test	1.53				

MRI	0.99
Muscle circumference	0.99
Skinfold thickness	0.99
NSRI	0.99
TTE	0.99
US	0.99
Sit and reach test	0.99
HR Max	0.66
1 leg balance test	0.66
Tinetti Gait and Balance Test	0.66
Lactate threshold	0.66
Chair stands	0.66
Lipids	0.66
Histology	0.66
Stair climbing	0.66
Waist to hip ratio	0.33
HR variability	0.33
Arm curl test	0.33
5-RM	0.33
Berg balance scale	0.33
Dynamic GSI	0.33
Heel rises	0.33
Static GSI	0.33
Augmentation index	0.33
Activities specific Balance Confidence Scale	0.33

Waist to hip ratio	1.53
5-RM	0.76
Anaerobic threshold	0.76
Arterial FMD	0.76
Barthel Index	0.76
Berg balance scale	0.76
Dynamic GSI	0.76
Functional reach	0.76
Gait speed	0.76
Graded exercise ergometer (time on test)	0.76
Heel rises	0.76
Histology	0.76
Incremental step test	0.76
Lipids	0.76
MIP/MEP	0.76
MRI	0.76
Muscle circumference	0.76
Muscle CSA	0.76
Stair climbing	0.76
Static GSI	0.76
Toe lifts	0.76

Mini-BEST	0.33
Resting muscle VO2	0.33
Toe-brachial index	0.33
Arterial stiffness index	0.33
2-min stair climbing	0.33
Squat test	0.33
SF-8	0.33
CIMT	0.33
Hamstring flexibility	0.33
Shoulder flexibility	0.33
FSST	0.33
4 min step test	0.33
Normal walking test	0.33
Fast walking test	0.33
20ft walk	0.33
Holter	0.33
FIM	0.33
Leg extensor power rig	0.33
SIP	0.33
Arterial baroreflex sensitivity	0.33
Arterial FMD	0.33
Leg extensor endurance	0.33
Forearm endurance	0.33
Spitzer index	0.33
Graded exercise ergometer (peak power)	0.33

Abbreviations: HD, hemodialysis; PD, peritoneal dialysis; KTx, kidney transplant; CKD, chronic kidney disease; Kidney disease quality of life (KDQOL), Short form 36 (SF-36), Short-form 8 (SF-8), Spitzer index, Sickness impact profile (SIP); HGS, hand grip strength; RM, repetition maximum; ADL, activities of daily living; FIM, functional independence measure; TUG, timed up and go; Mini-BEST, mini balance evaluation systems test; GSI, global stability index; 30s STS, 30 second sit to stand; 60s STS, 60 second sit to stand; 10 STS, 10 times sit to stand; 5 STS, 5 times sit to stand; FMD, flow-mediated dilation; CIMT, carotid intima-media thickness; HbA1c, glycosylated haemoglobin; OGTT, oral glucose tolerance test; VO_2, peak oxygen uptake; TTE, transthoracic echocardiography; HR, heart rate; 6MWT, 6-minute walk test; NSRI, North Staffordshire Royal Infirmary; ISWT, incremental shuttle walking test; METS, metabolic equivalents; BMI, body mass index; DEXA, dual x-ray absorptiometry; BIA, bioimpedance analysis; CT, computed tomography; CSA, cross-sectional area; US, ultrasound; MRI, magnetic resonance imaging; BMD, bone mineral density.

PRISMA Checklist:

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-13
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3, 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4, 5, 6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	PROSPERO CRD42019131999
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4, 5, 6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplement 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4, 5, 6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4, 5, 6

Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4, 5, 6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	N/A
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4, 5, 6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4, 5, 6

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4, 5, 6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6, 7, 8, 9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6, 7, 8, 9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6, 7, 8, 9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	6, 7, 8, 9
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9-13

Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	9-13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	13

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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